Thm. Suppose that f is analytic in $\mathbb{C}_\infty$, except for isolated singularities $z_1, \cdots, z_n, \infty$.
Then, $\operatorname{Res}(f,\infty) + \sum_{k=1}^{n} \operatorname{Res}(f, z_k) = 0$.

Proof. By the hypothesis, for some $\delta > 0$, a Laurent's representation

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

is valid in $\delta < |z| < \infty$. Specifically, one $\delta = \max\{|z_k| \mid k = 1, 2, \cdots, n\}$.
Choose $R > \delta$, and γ be a curve, $\{z \in \mathbb{C} \mid |z| = R\}$, the clockwise direction.
And, for $|z| = R$, the Laurent's representation of f converges uniformly, so that

$$\int_\gamma f(z)\,dz = \sum_{n=-\infty}^{\infty} a_n \cdot \int_\gamma z^n\,dz = -2\pi i \cdot a_{-1}.$$

So, define $\operatorname{Res}(f,\infty) = -a_{-1}$.

Moreover, $f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$ is analytic for $|z| > R$

iff $f\left(\frac{1}{z}\right) = \sum_{n=-\infty}^{\infty} a_n z^{-n}$ is analytic for $0 < |z| < \frac{1}{R}$ $\left(R < \left|\frac{1}{z}\right| \Leftrightarrow |z| < \frac{1}{R}\right)$.

So $a_{-1} = \operatorname{Res}\left(\frac{1}{z^2} \cdot f\left(\frac{1}{z}\right), 0\right)$ $\therefore$ Actually, just definition.

Example. Let $f(z) = 1/(z-3)\cdot(z^n - 1)$ $(n \in \mathbb{N})$. Evaluate $I = \frac{1}{2\pi i} \int_{|z|=2} f(z)\,dz$.
By the Residue Theorem, $I = \sum_{k=1}^{n} \operatorname{Res}\left(f, \exp\left(2\pi i \cdot \frac{k}{n}\right)\right) = -\operatorname{Res}(f,\infty) - \operatorname{Res}(f,3)$

Since $\operatorname{Res}(f,\infty) = -\operatorname{Res}\left(\frac{1}{z^2}\cdot f\left(\frac{1}{z}\right), 0\right) = -\operatorname{Res}\left(\frac{1}{z^2} \cdot \frac{z^{n+1}}{(1-z)(1-z^n)}, 0\right) = 0$,

because the rational function is analytic at $z = 0$.
Consequently,

$$I = -\operatorname{Res}(f,3) = -\lim_{z\to 3} \frac{1}{z^n - 1} = \frac{1}{1-3^n},$$

because f has simple pole at $z = 3$.